Benjamin (Ben) Whewell

AI/ML Scientist · Machine Learning Engineering

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Public version — phone number and email address withheld. Request the complete CV at ben-whewell.com/resume.html.

SUMMARY

AI/ML scientist and engineer with a PhD in computational engineering and eight years of research and production experience building machine learning systems for large-scale scientific computing. Owns the full lifecycle: data ingestion and feature engineering through training, evaluation, deployment into production codebases, and performance monitoring with hands-on user adoption. Current work spans LLM and retrieval-augmented generation (RAG) systems, reinforcement learning for sequential decision-making, and neural surrogates that cut compute cost within a prescribed error tolerance. Strong Python and C++ engineering on Linux, GPU, and HPC infrastructure.

TECHNICAL SKILLS

Languages: Python, C++, SQL, R, Cython, Bash

ML & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, Optuna, OpenCV; neural operators (DeepONet), autoencoders, surrogate modeling, hyperparameter optimization

LLMs & NLP: Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, RAG pipelines, vector databases, embeddings, Ollama (local inference), Claude Code, Codex, OpenCode, Azure AI Foundry, Amazon Bedrock, Pydantic, prompt design and evaluation

Reinforcement Learning: Gymnasium, Stable-Baselines3, Ray RLlib, PettingZoo, multi-agent RL, custom environment design, reward shaping

MLOps & Deployment: Docker, Kubernetes, MLflow, Weights & Biases, TensorBoard, DVC, CI/CD, model monitoring, feature stores, FastAPI, Flask, Django

Data & Infrastructure: NumPy, Pandas, PySpark, Hadoop, PostgreSQL, MySQL, SQLite, MongoDB, Snowflake, ETL; AWS, Azure, Git/GitHub, Linux/Unix, HPC and GPU clusters

EXPERIENCE

AI/ML Scientist — May 2026 – Present
Los Alamos National Laboratory — *Los Alamos, NM*

- Build and validate end-to-end ML pipelines — ingestion, preprocessing, feature engineering, training, evaluation — for large-scale multiphysics simulation workloads.
- Deploy models into production HPC software with performance monitoring, and drive adoption by supporting end users and making the case for the models to research and engineering groups.
- Develop LLM- and RAG-based assistants over domain-specific technical corpora, covering retrieval, grounding, and scoring answers against expert-labeled ground truth before release.
- Build frameworks for large physics codes with Claude Code, Codex, and OpenCode; run Claude on a government-hosted cloud platform.

Metropolis Postdoctoral Fellow — February 2025 – May 2026
Los Alamos National Laboratory — *Los Alamos, NM*

- Trained reinforcement learning agents for sequential decision-making that reduces error in numerical simulations, including custom environment design and reward engineering.
- Applied ML-driven optimization to energy-group structures for nuclear data, shrinking data volume while holding simulation accuracy within tolerance.
- Prototyped LLM and RAG applications for technical document search and simulation workflows, and evaluated local versus hosted inference trade-offs.
- Awarded the competitive Nicholas C. Metropolis Postdoctoral Fellowship; conference paper award winner at the 2025 ANS Mathematics and Computational Methods conference.

Graduate Student Researcher — September 2018 – January 2025
University of Notre Dame — *Notre Dame, IN*

- Designed and shipped an ML pipeline that replaces expensive linear-algebra kernels with learned surrogates (autoencoders, deep networks, deep operator networks), achieving data reduction in production simulations within a prescribed error tolerance.
- Built open-source scientific Python, Cython, and C++ libraries with published API documentation and regression tests, used by other researchers.

- Developed single-grid error estimation methodology for model-risk evaluation of high-dimensional solvers; published in a peer-reviewed verification and UQ journal.

Graduate Student Assistant June 2021 – August 2021
Lawrence Livermore National Laboratory *Livermore, CA*

- Integrated parallel C++ production physics codes across Livermore's HPC systems.
- Coordinated across multiple developer teams and downstream users to reconcile code-base and user requirements.

Graduate Student Assistant May 2019 – July 2019
Los Alamos National Laboratory *Los Alamos, NM*

- Built ML models to quantify experimental error and detect outliers in nuclear cross-section data using measurement features and covariances; results published in *Nuclear Instruments and Methods in Physics Research A*.

Boilers and Turbines Operator June 2015 – June 2018
NRG Energy Power Station *Middletown, CT*

- Operated and maintained safety-critical power-plant systems under strict procedures, with shift-handoff communication to supervision and plant personnel.

SELECTED OPEN-SOURCE PROJECTS

MyPDFLibrary — RAG command-line assistant over PDF corpora: local LLM inference with Ollama, embedding-based vector retrieval, and grounded question answering. `bwhewe-13/MyPDFLibrary`

WOLF — Debugging and reliability contributor to Los Alamos National Laboratory's open-source agent framework (Python, FastAPI, Chroma): async coroutine leak, BM25 empty-corpus crash, PyMuPDF deprecation cleanup, and a missing knowledge-base PDF ingestion endpoint, root-caused and patched. `lanl/wolf`

rl-lab — Reinforcement learning library and tutorial series spanning tabular, deep, multi-agent, and distributed RL with PyTorch, Gymnasium, Stable-Baselines3, and Ray RLlib. `bwhewe-13/rl-lab`

ants / discrete1 — Documented Python, Cython, and C++ simulation libraries with hooks for ML surrogate models and published API docs. `bwhewe-13/ants`

EDUCATION

Doctor of Philosophy, Mechanical and Aerospace Engineering January 2025
University of Notre Dame

Master of Science, Mechanical and Aerospace Engineering May 2022
University of Notre Dame

Bachelor of Science, Nuclear Engineering December 2017
Thomas Edison State University

SELECTED AI/ML PUBLICATIONS

- **B. Whewell**, R. G. McClarren, N. Gibson, and A. Khatiwada (2026). Application of Deep Operator Networks for Data Reduction in Deterministic Neutron Calculations. *Under review.*
- **B. Whewell**, N. Gibson, and A. Khatiwada (2026). Application of Reinforcement Learning for Multigroup Energy Grid Optimization for Neutron Transport Criticality Problems. *Under review.*
- **B. Whewell** and R. G. McClarren (2022). Data Reduction in Deterministic Neutron Transport Calculations Using Machine Learning. *Annals of Nuclear Energy*, 176, 109276. doi:10.1016/j.anucene.2022.109276
- **B. Whewell**, M. Grosskopf, and D. Neudecker (2020). Evaluating ^{239}Pu(n,f) cross sections via machine learning using experimental data, covariances, and measurement features. *Nuclear Instruments and Methods in Physics Research A*, 978, 164305. doi:10.1016/j.nima.2020.164305
- **B. Whewell** and R. G. McClarren (2025). Single Grid Error Estimation for Neutron Transport Solvers. *Journal of Verification, Validation and Uncertainty Quantification*, 10(2), 021001. doi:10.1115/1.4069426

Full record — 6 journal articles, 6 conference papers, 5 presentations, 2 technical reports: ben-whewell.com/research.html.

AWARDS & PROFESSIONAL SERVICE

- Nicholas C. Metropolis Postdoctoral Fellowship, Los Alamos National Laboratory.
- Conference Paper Award, 2025 ANS Mathematics and Computational Methods conference.
- Session Chair, Machine Learning and Artificial Intelligence I & IV, 2025 International Conference on Mathematics and Computational Methods Applied to Nuclear Science and Engineering.
- Reviewer, *Annals of Nuclear Energy* and ANS Mathematics and Computation Division (2025–present).
- Volunteer judge, Los Alamos National Laboratory 2025 Annual Student Symposium.